

PolarProtDb: A Database of Transmembrane and Secreted Proteins showing Apical-Basal Polarity

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Abstract

Most cells in multicellular organisms are somehow asymmetric, polarized: maintaining separate membrane domains. Typical examples are the epithelial cells (apical-basal polarization), neurons (dendritic-axonal domains), or migratory cells (with a leading and a trailing edge). Here we present the most comprehensive database containing experimentally verified mammalian proteins that display polarized sorting or secretion, focusing on epithelial polarity. In addition to the source cells or tissues, homology-based inferences and transmembrane topology (if applicable) are all provided. PolarProtDb also offers a detailed interface displaying all information that may be relevant for trafficking: including post-translational modifications (glycosylations and phosphorylations), known or predicted short linear motifs conserved across orthologs, as well as potential interaction partners. Data on polarized sorting has so far been scattered across myriads of publications, hence difficult to access. This information can help researchers in several areas, such as scanning for potential entry points of viral agents like COVID-19. PolarProtDb shall be a useful resource to design future experiments as well as for comparative analyses. The database is available at http://polarprotdb.enzim.hu.

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Introduction

Subcellular polarity is an essential feature of multicellular life.[1] Such polarity typically involves the formation of separate membrane domains on the same cell. These domains are created and actively maintained by a sophisticated sorting machinery, capable of separating transmembrane proteins destined for different membrane regions.[2] Eukaryotic proteins need to be precisely sorted after their synthesis at the endoplasmic reticulum (ER) or during their passage through the Golgi apparatus, and directed into separate vesicles en route to the cell surface.[3] Mislocalized proteins also have to be endocytosed from the plasma membrane, and then re-sorted to the correct domain.[4]

One of the most pervasive polarity phenomena in multicellular animals is observed in epithelial tissues.[5] Superficial cells need to maintain an apical membrane facing either the environment or internal cavities (lumina); as well as a basal - lateral surface that interfaces with other cells of the body. These two compartments are separated by tight junctions, where a cluster of adhesion proteins between adjacent epithelial cells prevents lateral diffusion of membrane proteins.[6] After the

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establishment of apico-basal polarity during cell differentiation, these two compartments are actively maintained in all epithelia. They possess very different adhesion proteins, transporters, and receptors and serve as a drop-off point for very different secretory proteins, enzymes, hormones, and cytokines.

Although apico-basal polarity has been studied extensively in the past two decades, many questions remain. Epithelial tissues employ quite complex sorting rules for apical proteins and only slightly better understood machinery for basolateral ones.[7] With the advent of high-throughput-studies on membrane and sorting polarity, now there is an ever increasing demand for comparability of results across separate experiments, methods, and tissues. The primary purpose of compiling PolarProtDb has been to provide a freely accessible, concise database of proteins with polarized sorting phenomena, focusing on the apico-basal separation of transmembrane and secreted proteins.

Numerous experiments have been reported in the past decades, from different model organisms and even well-established cell lines, yet a comparison of results is difficult and time consuming. Data from related proteins from different species are biologically especially valuable since they enable direct functional and evolutionary comparisons. The same is true for experimental sorting data from different tissues of the same organism since not all tissues handle the same protein in an identical way.[8] Yet as proteins of most mammalian species have rather similar sequences and often near-identical functions, they tend to possess similar polarity in the same cell type. While this can easily be ascertained by the most commonly studied organisms, the annotation can also be transferred to other mammals relatively safely. With the establishment of the right orthology groups, we can compare the conservation of function, the topology, and any sub-protein features associated with trafficking and sorting.

In order to understand polarized sorting, we have to reach sub-molecular levels: short linear motifs have been identified as vital for apical or basolateral localization, and post-translational modifications (especially specific glycosylation events) are just as important if not more than the motifs themselves. Although only a handful of these were identified in the past decades, the physiology of phosphorylation and glycosylation sites gets better described every year. Naturally, predicting the gross structure, folded domains, as well as the transmembrane topology are absolute necessities for any molecular studies. Thus proteins can only be truly compared with detailed alignments of related orthologs, with all domain, topology, motif, and modification data shown at the same time. In the case of N- and O-glycosylation, the link with apical sorting mechanisms justifies a thorough representation of not just the site, but the actual glycan structure.[9,10] The protein entries in PolarProtDb were designed to provide every bit of this information.

Results and Discussion

Articles and experiments incorporated into PolarProtDb

Using experimental data from 384 original research articles, we managed to incorporate 950 experiments into the database. For methods where multiple proteins were measured simultaneously, we regarded them as separate experiments. We did not discriminate against earlier results, if they were appropriately substantiated with the proper method. The processed articles span an impressive 34 years range, from 1987 to 2020, in a completely continuous fashion (we cite the largest number of individual papers and experiments from 2005, see Supplementary Figure 1). All included articles are presented with a PubMed and DOI link, for the ease of accession.

The layout of the database, presentation of orthologs, topologies, and post-translational modifications

The database provides various searching options and filters based on localization, tissue, experiment type, and more. After the user selects the protein of choice, a new site appears. On the top of the page, the gene name and the different experiments are displayed. To compare data from related proteins, a multiple sequence alignment of orthologs is shown in the middle, together with structural and functional information: It includes topology, Short Linear Motifs (SLiMs), phosphorylations and glycosylations. Users can zoom on the bars to enlarge it and view the individual sequences (Figure 1). At the bottom, region boundaries of all data for the particular protein are displayed, alongside other important data, such as motifs and binding partners.

Post-translational modifications (PTMs) are of key importance for the sorting of many membrane proteins. Hence we present them as part of the sequence and reference every database used as their source. Phosphorylation, in particular, was shown to be an important modulator of the subcellular trafficking of many proteins. A classic example is aquaporin 2 (AQP2), where a phosphorylation event is necessary for its incorporation into the apical membrane of renal collecting duct cells.[11] The gp130 subunit of interleukin 6 (IL6) receptor also carries a dileucine-type basolateral polarity and endocytosis signal, that is modulated by phosphorylation.[12,13] For our

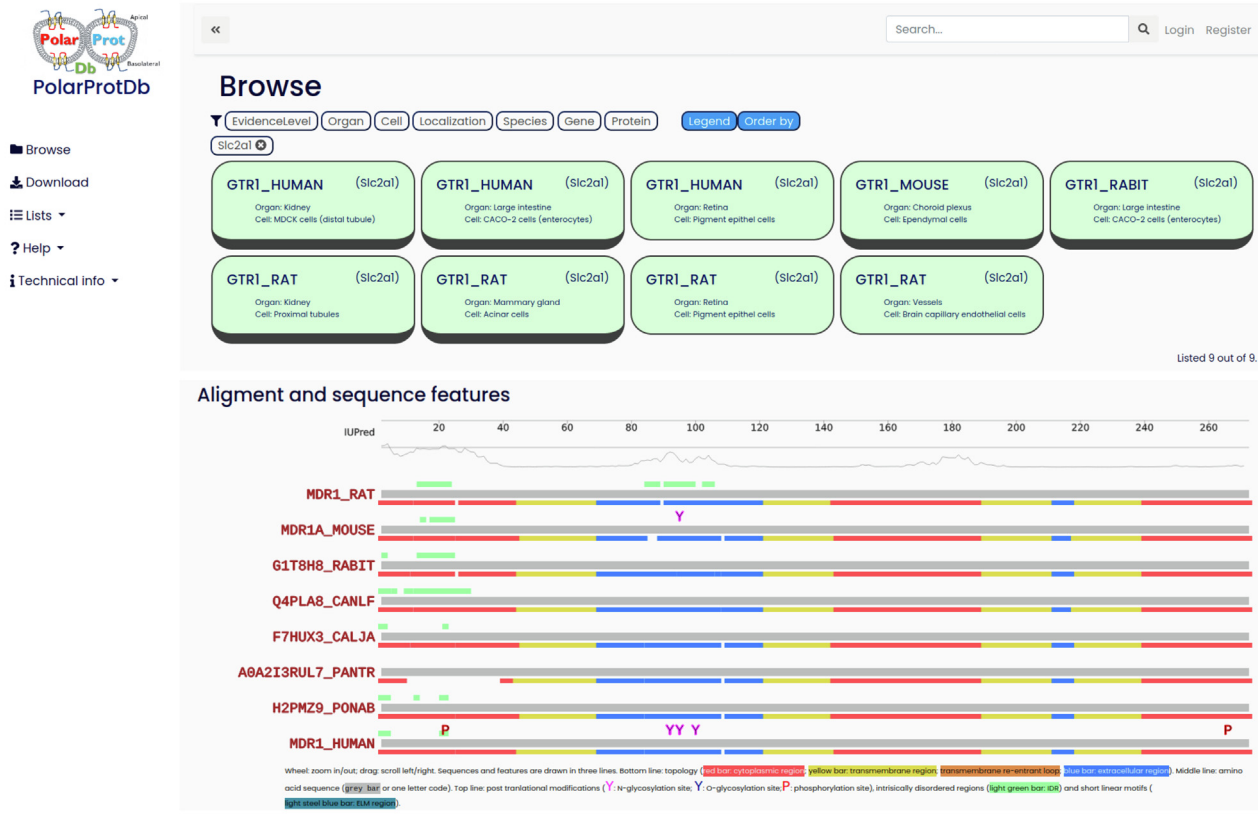


Figure 1. Snapshots from PolarProtDb, showing the side menu with the logo (left side), the gene list after searching for the mammalian GTR1 gene (right side, upper panel) and the display of topology and post-translational modifications (right side, lower panel) with colored bars and letters, respectively.

database, PhosphoSitePlus served as the source of phosphorylation sites.[27] They are displayed as small 'P' letters above each orthologous protein sequence, and the source can be viewed at the bottom of the page.

Glycosylation events represent another key player involved in protein sorting. Although not acting as a universal polarity signal, N-glycosylation events are important for proper polarized secretion for many proteins, such as erythropoietin (EPO) or WNT factors.[9,14] Similarly, O-glycosylation, especially mucin-type O-glycosylation can act as a strong apical sorting signal, for example, on mucin 1 (MUC1).[15] While the glycan density of a given protein surface is important, the structural characteristics of the most abundant glycoforms (N-glycan processing, terminal sialylation of O- and N-glycans, etc.) also play a role in polarized sorting.[9,10] Hence we also intended to collect detailed compositional and configurational information about the corresponding glycans. Nevertheless, exact site occupancy and the micro-heterogeneity of glycoforms are unfortunately not available for most known glycosylation sites. To supply the maximal information possible, we imported glycosylation patterns from a series of databases (see methods), and integrated all of them into the PolarProtDb database.

Protein-protein interactions and linear motifs involved in polarized sorting

Protein-protein interactions are of utmost importance for the proper localization of many proteins. To allow identification of potential piggybacking phenomena, we also integrated protein–protein interaction data to our site from IntAct,[16] viewable for each protein or orthology group separately. To help the analysis, experimentally validated interacting partners of a particular protein are also coloured with respect to their preferred localization. We estimated that in cases when both members of an interacting pair show polarized sorting to either side, they agree in localization 37 out of 48 (77%) cases, and display opposing polarity only 11 times (23%).

SLiMs are key mediators of protein–protein interactions, often directing or modulating protein subcellular sorting. Several of such trafficking motifs or potentially motif-containing short regions

Table 1 Short protein regions or linear motifs critical for polarized protein sorting according to experiments curated for the PolarProtDb database. Residues that are known or likely to be required for function, are bold and underlined.

Protein UniProt entry	Region or motif essential for **apical** polarity	Motif type	PubMed ID
MRP2_HUMAN	282-Q**S**QDA**L**VLE-290	Unknown	22454528
AT2B2_HUMAN	304-VKKGDGLQLPAADGAAASNAADSANASLVNG-334	Unknown	12624087
EAA3_HUMAN	501-KSYVNGGFAVD-511	Unknown	12486157
GTR3_HUMAN	473-DRSGKDGVMEMN-484	Unknown	14605095
GTR9_HUMAN-2	1-MKLSKKDRGEDEESDSAKKKL-21	Unknown	23361362
NPT2B_HUMAN	685-TEC**T**A**L**-690	PDZ-ligand	22904329
S22AB_HUMAN	545-VES**T**S**L**-550	PDZ-ligand	16236806
S22A5_MOUSE	552-LKS**T**A**F**-557	PDZ-ligand	16896066

Protein UniProt entry	Region or motif essential for **basolateral** polarity	Motif type	PubMed ID
ACM3_HUMAN	275-A**E**TEN**FV**-281	Dileucine-like	15870063
FURIN_HUMAN	782-G**E**RTA**FI**-788	Dileucine-like	10082580
IL6RB_HUMAN	781-**E**STQP**LL**-787	Dileucine-like	16274960
MICA_HUMAN	337-A**E**GPE**LV**-344	Dileucine-like	11854468
MOT3_HUMAN	476-L**E**ALE**VL**-482	Dileucine-like	21199217
MOT4_HUMAN	441-R**E**VEH**FL**-447	Dileucine-like	21199217
S23A2_HUMAN	51-N**E**DTE**LM**-57	Dileucine-like	23837633
S4A4_HUMAN	1052-M**E**QQP**FL**-1058	Dileucine-like	15273250
ST14_HUMAN	33-**EE**GVE**FL**-39	Dileucine-like	32049977
AREG_HUMAN	236-**EE**RKK**L**R-242	Monoleucine	21917092
BTC_HUMAN	156-**EE**MET**L**G-162	Monoleucine	26272915
SCF_HUMAN	254-N**E**ISM**L**Q-264	Monoleucine	11152680
TGFR1_HUMAN	161-**EE**DPS**L**D-167	Monoleucine	28768825
MOT3_HUMAN	455-EDA**E**A**E**GDSE**PL**P-467	Unknown	21199217
MOT4_HUMAN	422-AAA**EEE**KLHK**PP**A-434	Unknown	21199217
LRP1_HUMAN	4500-TNFT**NP**V**Y**ATL-4510	NPxY type	19005208
LRP1_HUMAN	4506-V**Y**AT**L**Y-4511	Tyrosine	19005208
CXAR_HUMAN	317-Q**Y**NQ**V**P-322	Tyrosine	22343291
EREG_HUMAN	155-E**Y**ER**V**T-160	Tyrosine	23671122
FCG2B_HUMAN	291-T**Y**SL**L**M-296	Tyrosine	8039492
FSHR_HUMAN	683-T**Y**IL**V**P-638	Tyrosine	9660834
FURIN_HUMAN	758-S**Y**KG**L**P-763	Tyrosine	10082580
KCNJ4_MOUSE	440-RRE**S**R**I**-445	PDZ-ligand	11504929
SDC1_MOUSE	306-QEE**F**Y**A**-311	PDZ-ligand	18764819

were identified in the curated articles (Table 1). To make them more concise, motifs that were clearly classifiable to a set of known trafficking/polarity motifs (PDZ domain ligands, dileucine or monoleucine-type AP1 binding motifs, tyrosine-based AP1 binding motifs or NPxY-type PTB domain ligands) were labelled accordingly. Unfortunately, most of them are not featured yet in public linear motif databases. In addition, there are many more motifs or experimentally identified motif-like regions that belong to yet-unidentified motif classes (the latter were labelled as “unknown motifs”). A complete set of applicable, known linear motifs (Supplementary Table 1) was also selected from the Eukaryotic Linear Motifs (ELM) database and automatically predicted for each protein.[28]

Classification of experimental data by cells types, tissues, and organs

Naturally, cell types were classified for easy viewing, based on the source organ. Still, some organs were split into multiple tissues due to didactic reasons (cornea and retina represent two very different sections of the eye). In contrast, others were grouped (especially tissues only present during early embryonic development). On the other hand, cell types that are known or suggested to show different sorting phenomena

were never pooled. One must still note that some cell types are considerably less studied than others. For example, few articles examined the immunofluorescent staining of goblet cells in the intestines, yet many reported the staining of enterocytes (that represent an easier to identify and much larger population). The same is true to proximal and distal tubules in kidney *versus* the alpha and beta intercalated cells. As noted before, many experiments were performed on cultured cell lines instead of complete organs. We listed a total of 20 different cell lines across all articles and also assigned these cell lines to their source organ for easier comparisons.

We feature 59 different mammalian cell types from 35 organs (not counting cell lines). The name of cell types tends to copy their anatomically most accepted name. As distant organisms can possess orthologous proteins of different functions and different localization, we generally refrained from including data on non-vertebrate proteins (e.g. from *D. melanogaster* or *C. elegans*). Proper alignment of such sequences is also highly problematic, especially with transmembrane or disordered segments. Similarly, the direct aligmnent of linear motifs could not be reliably performed on distantly related orthologs. Since functional data on transmembrane proteins is sparse on fishes, amphibians, or sauropsids, they were chosen not to be included in this database version. However, orthologous protein entries from distant species can still be viewed at our website (and Supplementary Table 7).

We are quite aware of the fact that not all tissues sort the same protein the same way. For example, while most epithelial tissues contain the K^+/Na^+ ATPase strictly at their basolateral membrane, choroid plexus cells possess them at their apical surface.[7] The reason behind these tissue- and protein-specific differences have only been clarified in a few cases. Kidney proximal tubule cells or the LLC-PK1 cell line, for example, do not express the basolaterally sorting adaptin complex AP-1B. As a result, they tend to sort several proteins like the transferrin receptor or LDL receptor to their apical surface, instead of sorting them basolaterally, as most other epithelial tissues do.[8] To account for these differences, we always collected experimental data by not just organisms or tissues/organs, but down to the actual cell type involved. As immortalized cell lines (when forced to differentiate) generally tend to preserve the sorting machinery and polarities of their source tissue, they are listed as part of the original organ. Therefore we consider LLC-PK1 cells as the pig kidney proximal tubule cell type, and MDCK cells as dog kidney distal tubule cells, while still noting the cell line.

In line with the above observations, a minor but still notable portion of proteins in PolarProtDb are sorted differently depending on the cell type. In the current version, there are 44 mammalian genes that encode proteins with such variable sorting (approx. 7% of all genes included). Since apical-basal polarity exists as a continuum, there are obviously some proteins that might appear polarized under one condition and less so under another: 23 (52%) of the above-cited anomalous examples appear to behave so. However, we also found 18 proteins (41%) that display a complete flip-flop of polarity in certain tissues, without ever appearing non-polar. Such proteins include the KCNQ1 ion channel,[17,18] Carbonic anhydrase 12,[19] and the multidrug-resistance transporter ABCB1.[20] Some examples (such as the Na^+ / K^+ ATPase or the LDL and transferrin receptors) are consistently confirmed by multiple articles,[8] they support the notion that the polarity flip is not merely a random experimental noise, but a true biological phenomenon. Highly complex polarization behaviour (seen as apical, non-polar, or basolateral in different tissues, respectively) also seems to exist, although very rarely (with only three examples in the database). All proteins with such complex sorting are listed on the website. For the gene-level comparisons across all mammals, see Supplementary Tables 3–6.

Presentation of experiments

Experimental methods were classified and grouped according to the scheme shown in Table 2, to allow easy viewing. While curating, we paid special attention to include data from all types of experiments. If an article published information about the motifs or sub-protein segments critically involved in protein localization, we usually indicated that by adding the proper label (sub-protein manipulation) to the method. A total of 5 high-throughput datasets were also curated for inclusion. In the latter, proteins were regarded as polarized if they showed at least 60–80% selectivity on either the apical or basolateral side (or 3–5 fold enrichment, based on sufficient data points). In the same high throughput articles, proteins that fell into the utterly indecisive 40%-60% range were regarded as non-polar ones. For low-throughput methods and protein-specific articles (antibody labelling, western blots, etc), we generally stuck to the judgement of the author(s) on polarity. The author statement on protein polarity was not accepted without experimental evidence presented.

Accessibility of the database and user interactions

PolarProtDb is open access, free online bioinformatics resource, under the address http://polarprotdb.enzim.hu/. It is designed to be able to accommodate a continuous stream of newly curated data. Volunteers are encouraged to register on the website, in order to be able to

Table 2 Classification of experimental methods used to assess protein polarity and their assignment to distinct evidence classes in PolarProtDb.

Experimental procedures	Experimental evidence type
Immunfluorescence (confocal microscopy)	Antibody staning
Immunogold labelling (electron microscopy)	
Confocal microscopy of GFP-tagged proteins	Fluorescent labelling
Isolation of of brush border and basal membranes	Membrane preparation
Side-specific release of membrane proteins with enzyme	
Side-specific biotinylation of proteins followed by purification and/or western blots	
Western blot of proteins in apical or basolateral media	Selective protein analysis
Immunoprecipitation of secreted proteins in either media	
Protease, nuclease or other enzymatic assay *in situ* with chromogenic substrate	Functional assay
Uptake or transport assays of specific substrates	
Mass-spectrometry of side-specifically labelled endogenous proteins	Large-scale proteomics
Comparison of specifically designed, localization-defective mutans with WT proteins	Sub-protein manipulation
Co-expression or co-purification experiments	Other (specified in notes)
Viral infection susceptability assays	

upload new protein candidates for apical-basolateral polarity (or lack thereof). The uploaded candidates will be revised by the PolarProtDb team and will be added to an ever growing pool of specifically sorted proteins. We expect PolarProtDb to be particularly useful for researchers focusing on protein trafficking and intracellular sorting.

Methods

Webserver description

Data on the webserver is stored in a relational database (MySQL, 8.0.4), the engine for processing user requests was written in Laravel framework (7.22.2) using Livewire (1.3) and Vue.js (2.5.17) with D3.js (5.16).

Literature curation

To obtain publications, we thoroughly searched PubMed, Google, and Google Scholar using keywords related to “apical”, “basolateral” and “polarized protein sorting”. All UniProt entries previously annotated for “basolateral plasma membrane” and “apical plasma membrane” were also checked for the publications linked. No automated annotation or review-based common knowledge was accepted as the base for inclusion into the database. All proteins must have had clear experimental proof for either apical or basolateral polarity (or lack thereof) for inclusion. Conflicting experimental results were intentionally not resolved (both contradicting articles were included) as long as their conclusions were not suspected to represent an experimental error.

Genes, proteins, and aligments

Based on the articles curated, we manually selected 604 proteins from 450 mammalian genes and assigned them to UniProt entries.[21] As a next step, we searched for orthologs in *Homo Sapiens*, *Pan Troglodytes*, *Gorilla Gorilla*, *Pongo Abelii*, *Macaca Mulatta*, *Felis Catus*, *Canis Familiaris*, *Equus Caballus*, *Ovis Aries*, *Bos Taurus*, *Oryctolagus Cuniculus*, *Callithrix Jacchus*, *Mus Musculus*, *Rattus Norvegicus*, and *Sus Scrofa* in the OMA database.[22] Orthologs were drawn from a standardized list, but they are allowed to be overridden if a manually curated better ortholog was provided with the experimental data. We discriminated TM/non-TM proteins and predicted TM topology using CCTOP.[23] We used ClustalΩ,[24] Muscle[25] and Praline2[26] – and used the topology input to enhance the alignment in the case of Praline. We selected the alignment with the best quality considering TM topology. However, there were still cases when the alignment algorithms produced erroneous output – in this case we discarded sequences that caused contradiction.

Linear motifs, secondary modifications, and interactome data

Experimentally-identified motifs and motif-like regions that were identified in the publications were extracted and re-grouped according to generic motif patterns already established for polarized sorting.[7] Phosphorylation data was extracted directly from PhosphoSitePlus (version 6.5.9.3).[27] Protein-protein interactions were retrieved from IntAct by filtering out spoke and matrix expanded co-complexes.[16] For motif predictions, we selected 38 linear motifs from the Eukaryotic Linear Motif database[28] that may play a role in sorting (Supplementary Table 1) and searched these motifs based on their regular expression. Intrinsically disordered regions (IDRs) were predicted using IUPred2a.[29] We only accepted motifs predicted in IDRs, when the motif occupied the proper localization (cytosolic/non-cytosolic) and when it was present across all aligned orthologs (these rules did not apply to experimental motifs).

Sources of glycosylation data

Due to the lack of comprehensive databases of N- and O-linked glycosylation, the corresponding sites, glycostructures, and monosaccharide compositions were imported from the GlyGEN[30] data integration platform extended with several medium-sized sources (Supplementary Table 2). For the investigation of carbohydrate fragments in RCSB PDB entries, the GlycoNAVI/Carbohydrate Structures in PDB(CSP) database from GlyCosmos[31] portal was utilized, with some corrections (removing both random and systematic errors). In addition, PDB entries representing polypeptide chains expressed in non-vertebrate systems were discarded from the analysis. In all cases, the UniProtKB-PDB mapping was performed by BLAST searches, and ClustalΩ was used to globally align the sequence pairs. For the identification of individual glycans and to represent their

structural characteristics, their GlyTouCan ID-s[32] or their names in WURCS [33] were used. Glycosylation sites lying in the intracellular domain of transmembrane proteins (ca. 3%, according to the predicted membrane topologies) were discarded, because they probably represent experimental or evaluation errors.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jmb.2020.11.004.

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